

Conditional Statements - Day 2 Assignment

Q1. What is a conditional statement in programming?

A conditional statement is used to make decisions based on conditions.

Q2. What does an if statement do?

It executes code when a condition is true.

Q3. What is the purpose of an else block?

It executes code when the if condition is false.

Q4. When is an else block executed?

When the if condition is false.

Q5. What is a Boolean expression?

An expression that evaluates to true or false.

Q6. Write the syntax of a simple if statement.

```
if(condition){ // code }
```

Q7. What operator is commonly used to test equality?

`==`

Q8. What does the modulus (%) operator return?

The remainder after division.

Q9. How can you check if a number is even?

```
if(num % 2 == 0)
```

Q10. How can you check if a number is odd?

```
if(num % 2 != 0)
```

Q11. What is the purpose of else-if?

To check multiple conditions.

Q12. What is a nested if statement?

An if statement inside another if statement.

Q13. Can an if statement exist without an else?

Yes.

Q14. What is the output when a false condition is tested in an if statement without else?

No output; the block is skipped.

Q15. What is the role of braces {} in C/C++ conditionals?

They define a block of code.

Q16. What does > mean?

Greater than.

Q17. What does < mean?

Less than.

Q18. What does >= mean?

Greater than or equal to.

Q19. What does <= mean?

Less than or equal to.

Q20. What does != mean?

Not equal to.

Q21. What is a switch statement?

A multi-way decision statement.

Q22. What is a case label in a switch statement?

A value checked by switch.

Q23. Why is break used in switch?

To exit the switch.

Q24. What happens if break is omitted?

Fall-through occurs.

Q25. What is the default case in switch?

Runs when no case matches.

Q26. Can switch compare strings in C++?

No.

Q27. Which statement is better for many constant choices: if-else or switch?

Switch.

Q28. How do you test whether a value is positive?

`if(num > 0)`

Q29. How do you test whether a value is negative?

`if(num < 0)`

Q30. How do you test whether a value is zero?

`if(num == 0)`

Q31. Write a condition to check if age is at least 18.

`if(age >= 18)`

Q32. Write a condition to check if marks are above 50.

`if(marks > 50)`

Q33. What is decision making in programming?

Choosing actions based on conditions.

Q34. Can nested if statements have multiple levels?

Yes.

Q35. What is logical AND (&&)?

True only if all conditions are true.

Q36. What is logical OR (||)?

True if at least one condition is true.

Q37. What is logical NOT (!)?

Reverses a Boolean value.

Q38. How do you check if a year is divisible by 4?

`if(year % 4 == 0)`

Q39. What is a leap year?

A year with 366 days.

Q40. What is the first condition for a leap year?

The year must be divisible by 4.

Q41. How do you find the largest of two numbers?

if(a > b)

Q42. How do you find the largest of three numbers?

if(a > b && a > c)

Q43. What is an if-else-if ladder?

A chain of conditions.

Q44. What is the difference between if and switch?

If handles complex conditions; switch handles fixed values.

Q45. Can switch use ranges directly?

No.

Q46. What is fall-through in switch?

Execution continues to the next case.

Q47. What data type is usually used for conditions?

bool

Q48. Why are conditional statements important?

They help programs make decisions.

Q49. Give one real-life example of a conditional statement.

If the light is green, move; otherwise stop.

Q50. Name four conditional structures covered.

if, if-else, else-if ladder, switch